

Executive Summary — Group-Split Data Leakage Fix

Issue #11 | XGBoost Binary Classifier | Data: 2015–2024 | April 11, 2026

BACKGROUND

The classifier predicts pup genotype (**HET** vs **WT**) from USV syllable features. Each of the **87 mice** contributes 6–337 recordings (avg ~109). The previous baseline split recordings randomly — syllables from the **same mouse** appeared in both train and test sets, inflating reported accuracy.

Data Leakage Confirmed

In the baseline split, **87 out of 87** mice are shared between train and test. The model memorized mouse-specific patterns rather than learning generalizable ASD features.

WHAT CHANGED

Added `--group-split` flag to `train_classifier.py`. When active, the split ensures **no mouse appears in more than one set** (train / val / test). Mice are split with stratification on genotype label, then rows are selected by mouse membership.

Baseline Split (random, leaky)

Train: 5,709 rows | Val: 1,903 | Test: 1,903

Mouse overlap: **87/87** shared across sets

Group Split (no leakage)

Train: 5,593 rows (51 mice) | Val: 2,020 (18 mice) | Test: 1,902 (18 mice)

Mouse overlap: **0** — fully disjoint

RESULTS COMPARISON

72.1%

Test Accuracy

▼ -10.8% from baseline

0.73

Weighted F1

▼ -0.10 from baseline

88.3%

Train Accuracy

▲ +0.9% (no change)

Metric	Baseline (leaky)	Group Split	Change
Test Accuracy	82.9%	72.1%	-10.8%
Weighted F1	0.83	0.73	-0.10
HET Precision	0.73	0.53	-0.20
HET Recall	0.90	0.80	-0.10
HET F1	0.81	0.64	-0.17
WT Precision	0.92	0.89	-0.03
WT Recall	0.78	0.69	-0.09
WT F1	0.84	0.77	-0.07
Train Accuracy	87.4%	88.3%	+0.9%

WHY DID PRECISION DROP?

1. The model relied heavily on mother_gen

In the baseline, mother_gen contributed **57%** of total feature importance. All HET mice have mother_gen=0, but many WT mice also have mother_gen=0. With leakage, the model could distinguish them by memorizing individual mice. Without leakage, it sees mother_gen=0 and over-predicts HET — producing **false positives** that drop precision from 0.73 to **0.53**.

2. Imbalanced test set

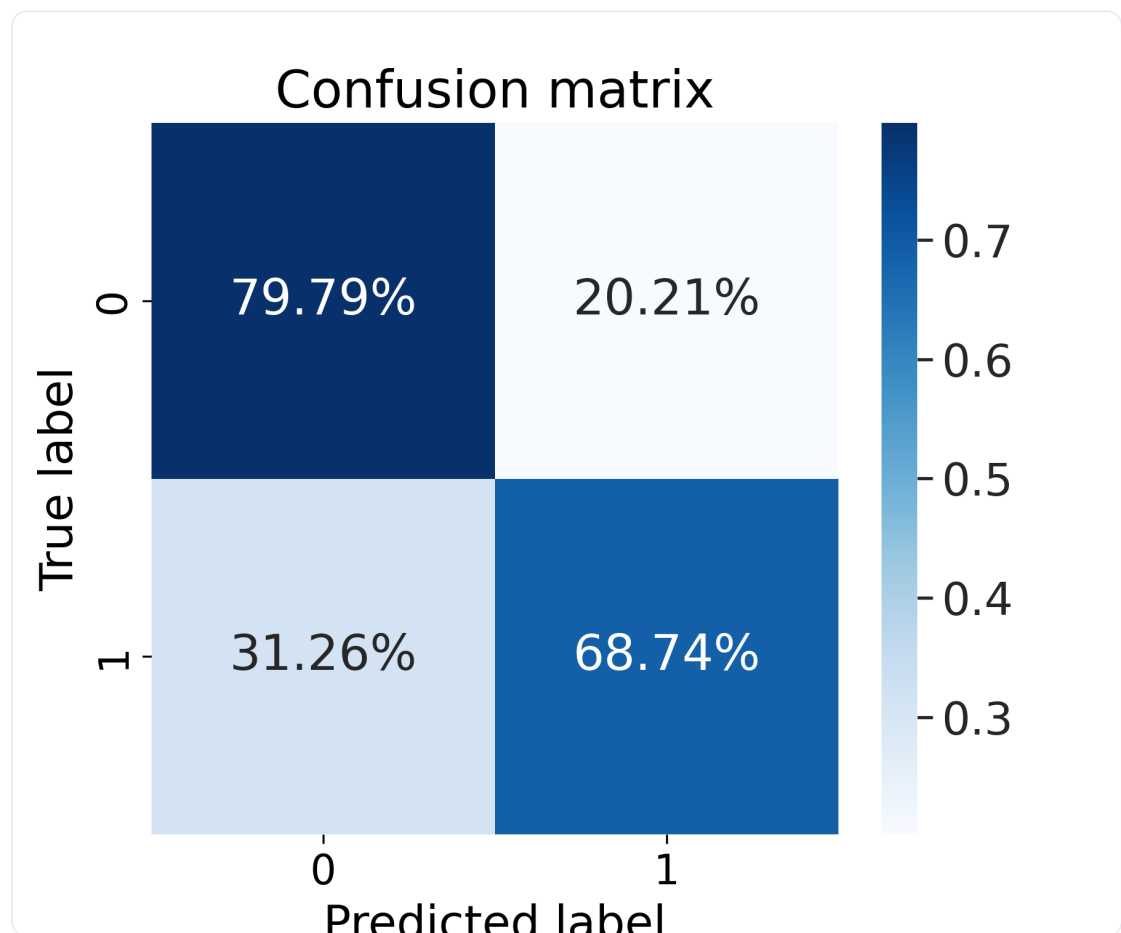
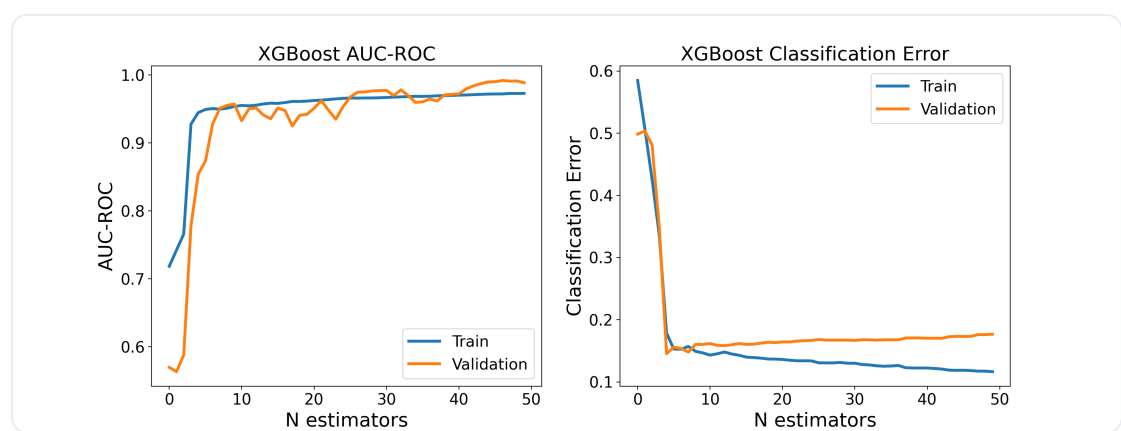
The group split produced a test set with **30.7% HET / 69.3% WT** (7 vs 11 mice). With fewer HET samples and more WT samples, any false positive weighs heavier against HET precision.

3. Strain distribution mismatch

Train has 63% Strain 1 but test has only 32% Strain 1. The model trains on one distribution and is tested on another, reducing generalization.

AUC-ROC VS ACCURACY PARADOX

The AUC-ROC on validation reaches **~0.97**, yet test accuracy is only 72.1%. This means the model **ranks** mice correctly (HET gets higher probability than WT), but the decision **threshold** (0.5) is not calibrated. The model "knows" who is HET, but the cutoff is wrong.



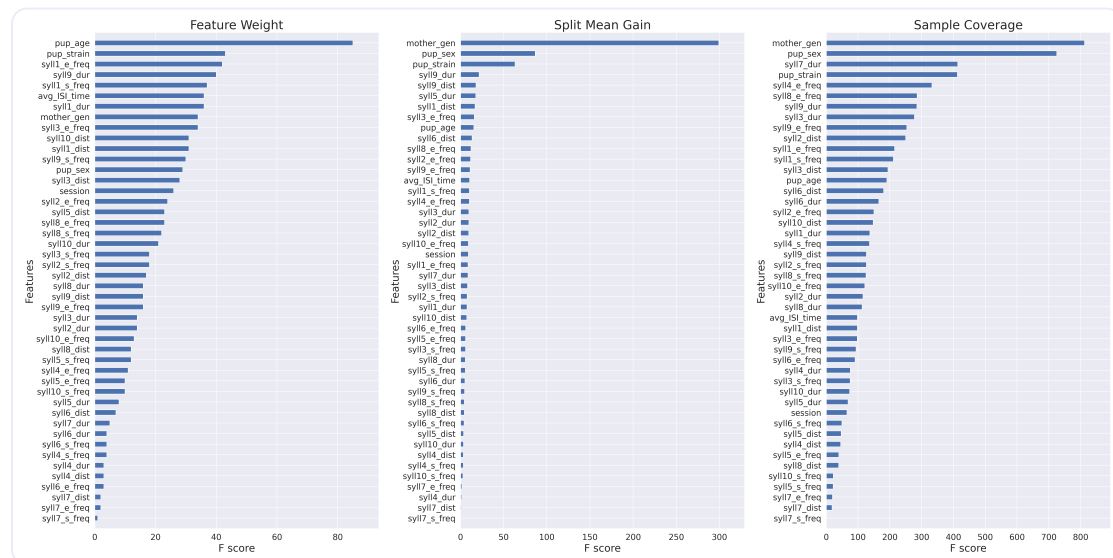
The Classification Error plot also reveals **overfitting**: after ~10 estimators, validation error rises while train error continues to fall. The current model uses 50 estimators — far past the optimal point.

CONFUSION MATRIX

	Predicted HET	Predicted WT
Actual HET	466	118
Actual WT	412	906

The core problem: **412 false positives** — WT mice wrongly classified as HET. This is what drives HET precision down to 0.53.

FEATURE IMPORTANCE



`mother_gen` remains the dominant feature at **37%** importance (down from 57% in baseline). Acoustic syllable features gained relative weight, which is a positive sign — the model is starting to use actual vocalization patterns.

Key Takeaway

The baseline accuracy of 82.9% was **inflated by ~10 percentage points** due to data leakage. The real generalization performance is **72.1%**. This is now the honest baseline for future improvements — any gain from here is genuine.

RECOMMENDATIONS FOR NEXT STEPS

- **Threshold tuning** — AUC is 0.97, meaning the model ranks well. Calibrating the decision threshold (instead of default 0.5) could significantly improve precision without retraining.
- **Reduce estimators / early stopping** — The error curves show overfitting after ~10 estimators. Reducing `n_estimators` to 10–15 or adding early stopping on validation error may improve generalization.
- **Remove or reduce `mother_gen`** — It contributes 37% of importance but leaks genetic information that the model over-relies on. Training without it would force the model to learn from acoustic features.

- **Balance strain in split** — Train has 63% Strain 1 while test has 32%. A strain-balanced group split could reduce distribution mismatch.
- **Group by mother (not just pup)** — Siblings share genetics. Grouping by mother identity would prevent sibling leakage. Requires adding mother ID to the feature pipeline.

USAGE

Command	Description
<code>python3 src/classification/train_classifier.py</code>	Baseline (random split, same as before)
<code>python3 src/classification/train_classifier.py --group-split</code>	Group-aware split (no leakage)
<code>python3 src/classification/train_classifier.py --group-split --results-dir my_dir</code>	Custom output directory